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Quiz 7

Difference:

1. Gradient Space

1. Policy Gradient: Updates in the Euclidean parameter space
2. Natural Policy Gradient: Updates in the policy probability distribution space considering distribution geometry

2. Update Rule

1. Policy Gradient: $\theta \leftarrow \theta + \alpha \nabla_\theta J(\pi_\theta)$
2. Natural Policy Gradient: $\theta \leftarrow \theta + \alpha \mathbf{F}^{-1} \nabla_\theta J(\pi_\theta)$, where $\mathbf{F}$ is the Fisher information matrix

3. Step Property

1. Policy Gradient: Parameter steps do not guarantee consistent policy changes
2. Natural Policy Gradient: Ensures updates with constant KL divergence, leading to stable policy changes

4. Stability and Efficiency

1. Policy Gradient: Sensitive to parameter scaling, unstable and slow
2. Natural Policy Gradient: Parameter-invariant, more stable and efficient

5. Computation Cost

1. Policy Gradient: Low computation cost
2. Natural Policy Gradient: High cost due to Fisher matrix inversion